

From the Editor's Desk

The present issue of *Prajnanabha: Information and Physics* brings together a diverse yet conceptually unified collection of works that collectively advance what may be termed a *post-Shannon* paradigm of information theory and its interdisciplinary extensions. The contributions span foundational developments in information theory, quantum foundations, biological information processing, and geometric-combinatorial constraints arising in natural systems. Despite their apparent diversity, they are linked by a central theme: the recognition and exploitation of *structure*—whether in probability spaces, error events, biological architectures, or informational substrates.

We begin with the lead article on *Post-Shannon Information Theory*, which lays the conceptual groundwork for the entire issue. This work challenges the classical binary abstraction of typical versus atypical events by introducing a refined geometric perspective in which atypical and error sets possess internal structure. By partitioning these sets into clusters, the author introduces a tunable resolution parameter that interpolates between classical Shannon theory and a richer, structure-aware regime. This perspective is not merely philosophical; it yields concrete refinements in rate–distortion and reliability tradeoffs, opening the door to improved finite-blocklength performance and new operational interpretations.

Building on this foundation, the subsequent paper on *Structured Source Coding* provides a rigorous and quantitative realization of these ideas. By clustering the atypical set using distortion metrics such as Hamming distance, the work derives refined bounds on dominant error probabilities and demonstrates, both theoretically and via simulation, that the internal geometry of the source can be exploited for improved performance. Importantly, the classical Shannon result is recovered as a limiting case, ensuring consistency while extending applicability.

Complementing this, the article on *Structured Channel Coding* extends the structured paradigm to communication channels, particularly in the context of Gaussian feedback systems. Here, the monolithic decoding-error event is decomposed into clusters, and reliability is redefined in terms of dominant error modes. This leads to a structured reliability function that surpasses classical bounds under the refined metric. The work is notable for its careful treatment of both achievability and converse bounds, ensuring that the proposed gains are grounded in rigorous analysis.

The issue then transitions to foundational questions in quantum theory with the article on *Nonlocal Unification as the Completion of QBism*. This work proposes an ontological extension of QBism by introducing a universal informational substrate from which spacetime and physical observables emerge via agent-based sampling. By grounding agents in biological reality and framing measurement as a sampling process, the paper bridges quantum foundations, information theory, and cognitive science. It offers a compelling framework in which subjectivity, nonlocality, and emergence coexist within a unified informational paradigm.

In a different yet equally insightful direction, the paper on *Kekeya Upper Bounds for Neural Arbors* establishes a striking connection between incidence geometry and neurobiology. By modeling neural branching structures as bounded-degree incidence graphs, the work demonstrates that biological constraints such as directionality and acyclicity enforce linear bounds

on complexity, in contrast to the superlinear bounds achievable in unconstrained geometric settings. This result highlights a fundamental gap between mathematical possibility and biological realizability, with implications for understanding the architecture of neural systems.

The issue concludes with a set of earlier arXiv contributions that are included for completeness, wider dissemination, and archival purposes. These works, forming the final four entries of the issue, explore themes such as information processing in biological systems and quantum computational models of the brain. Their inclusion ensures that foundational ideas developed over time are preserved within a unified narrative and made accessible to a broader audience.

Taken together, the contributions in this issue reflect a shift from coarse, monolithic descriptions toward finely resolved, structure-aware frameworks. Whether in coding theory, quantum foundations, or biological systems, the recognition of internal structure emerges as a unifying principle with both theoretical and practical significance. It is our hope that this issue stimulates further exploration of these ideas and fosters new connections across disciplines.

Editor